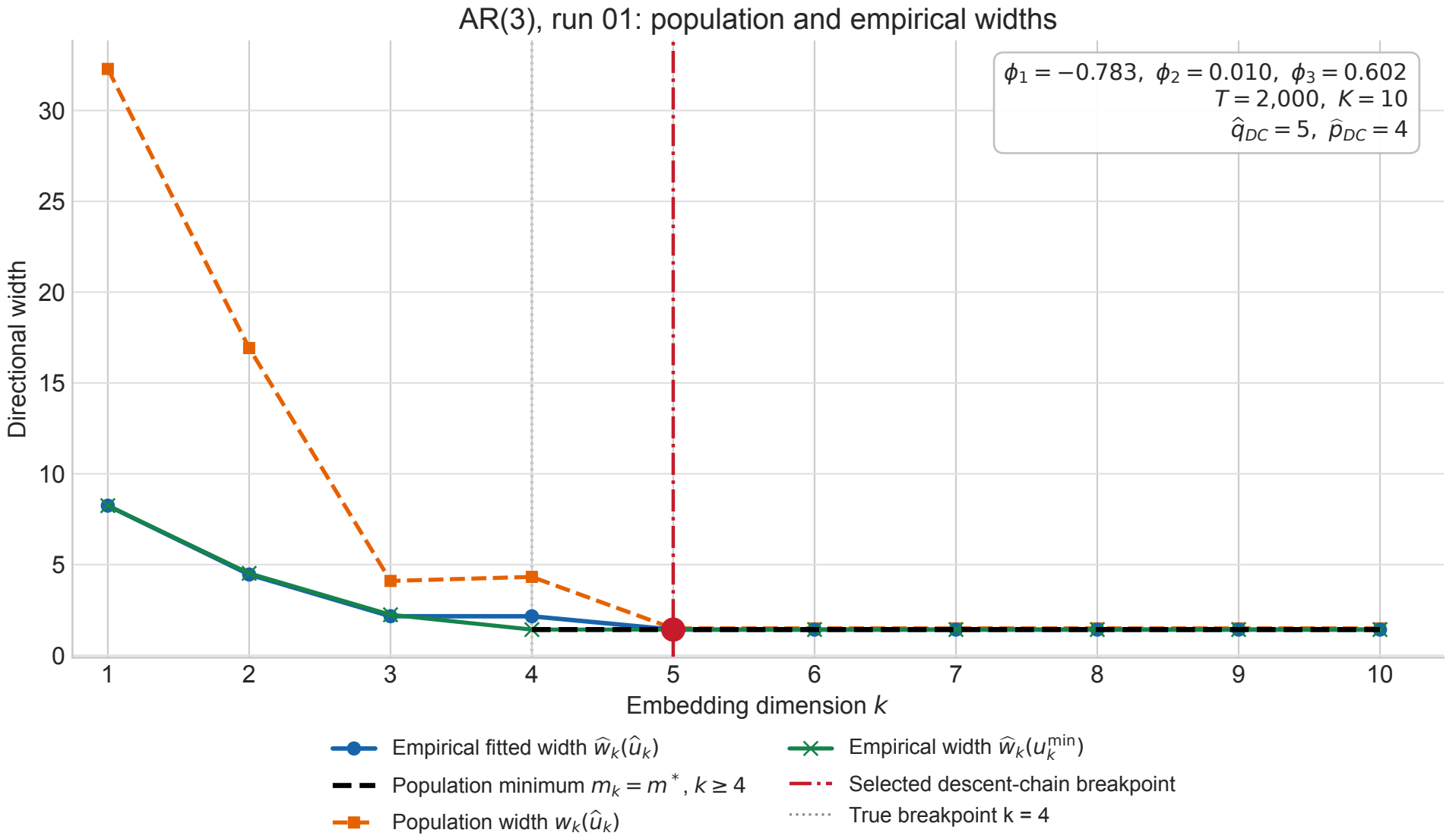
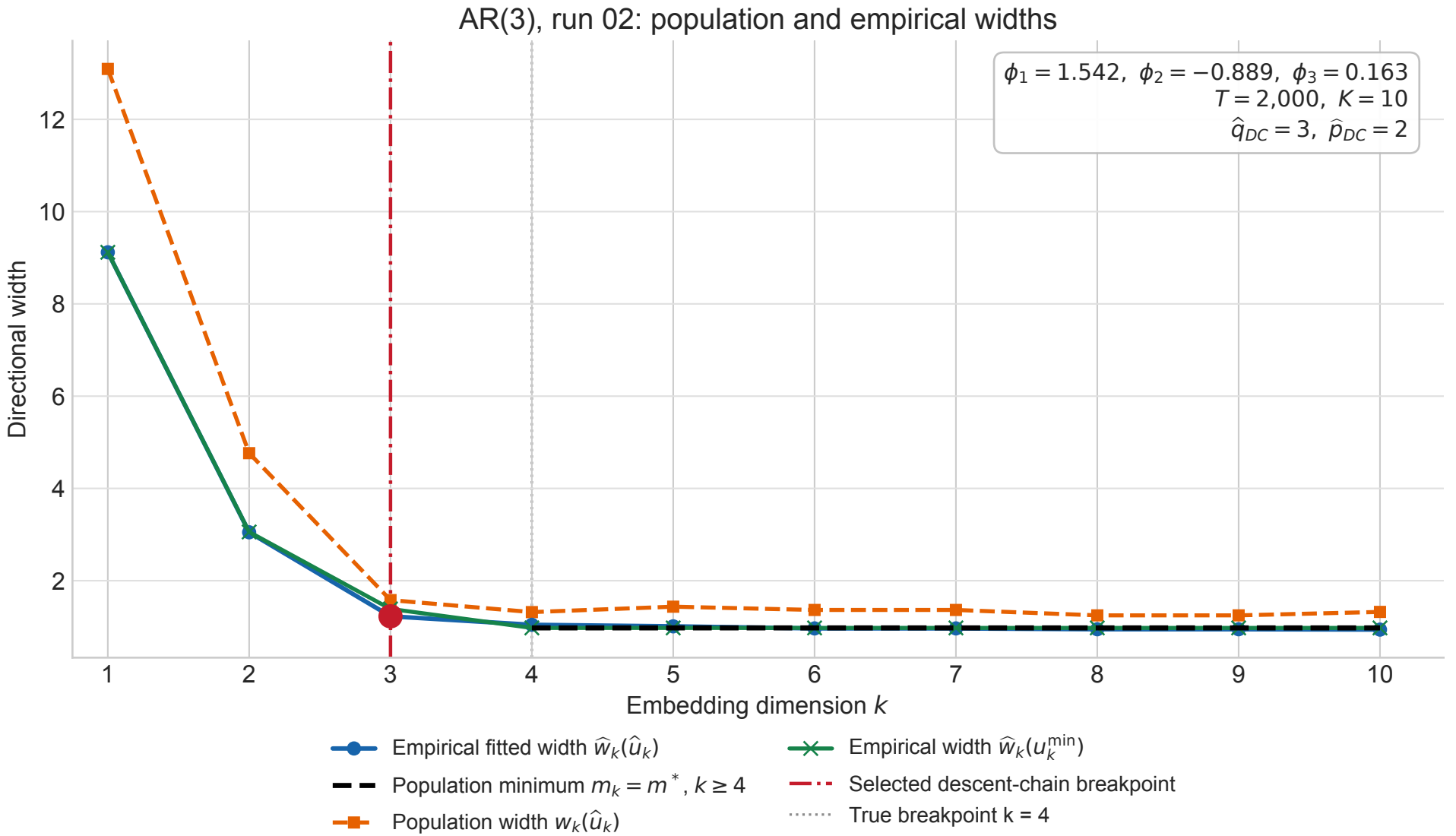
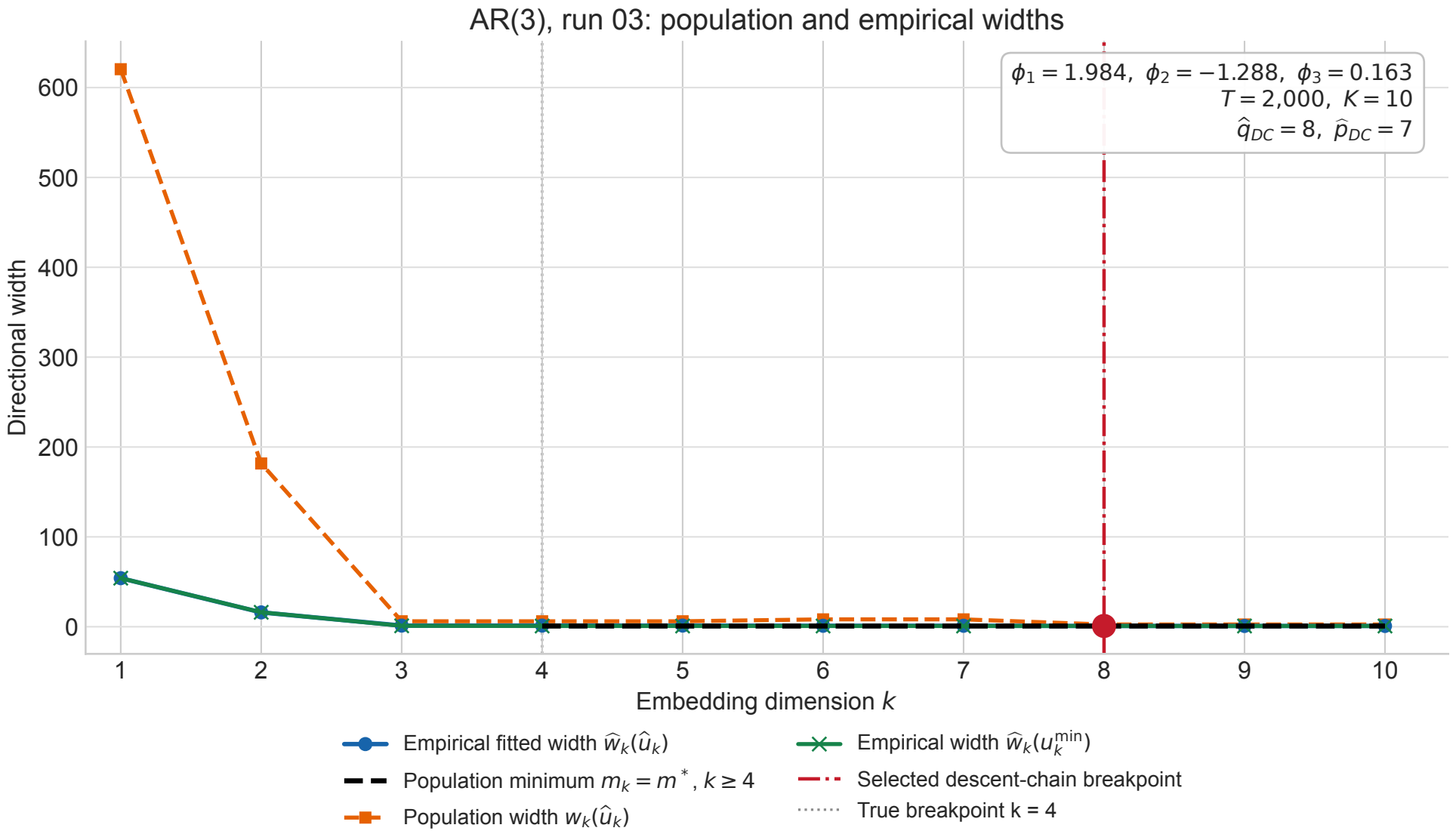
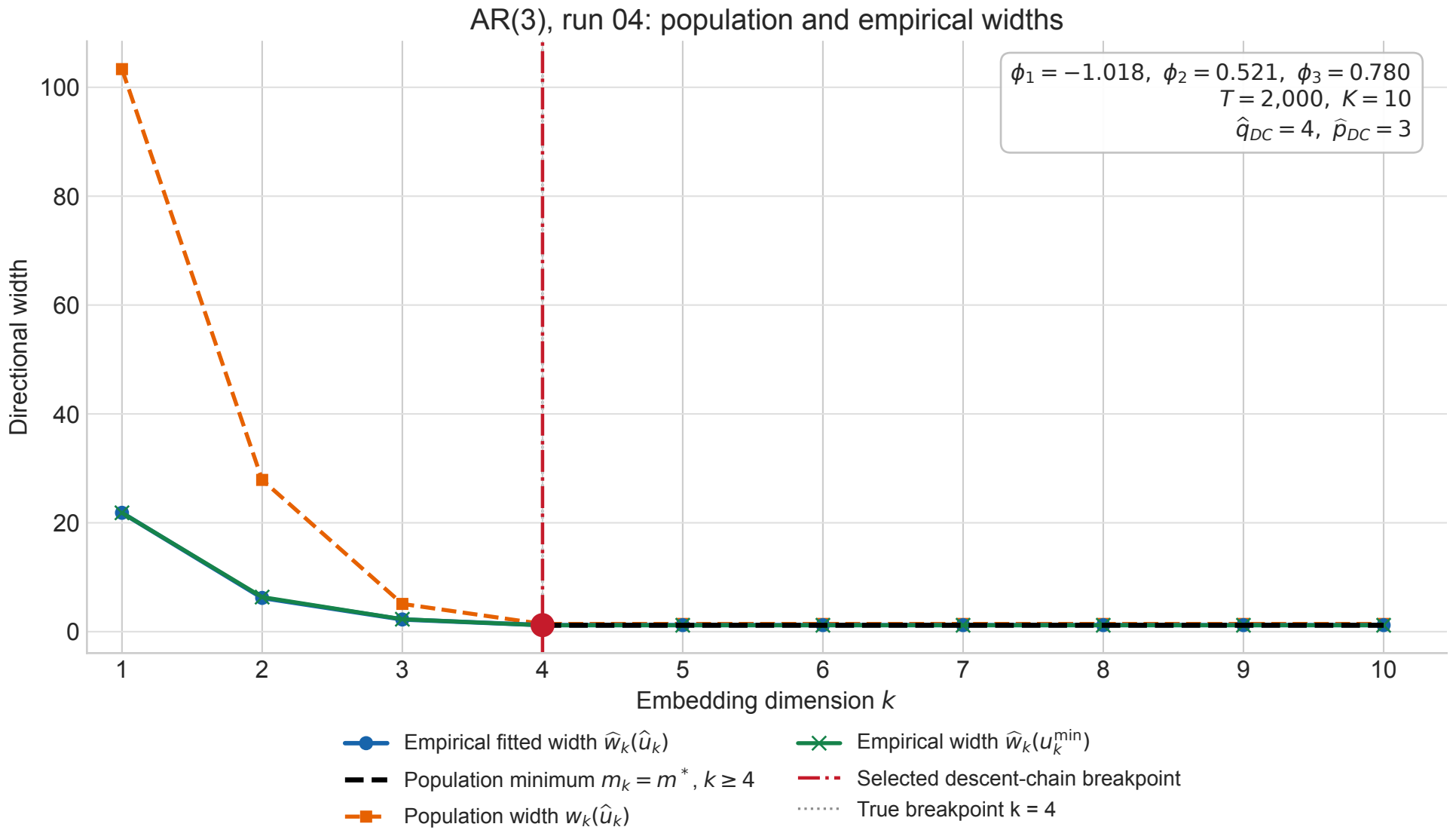
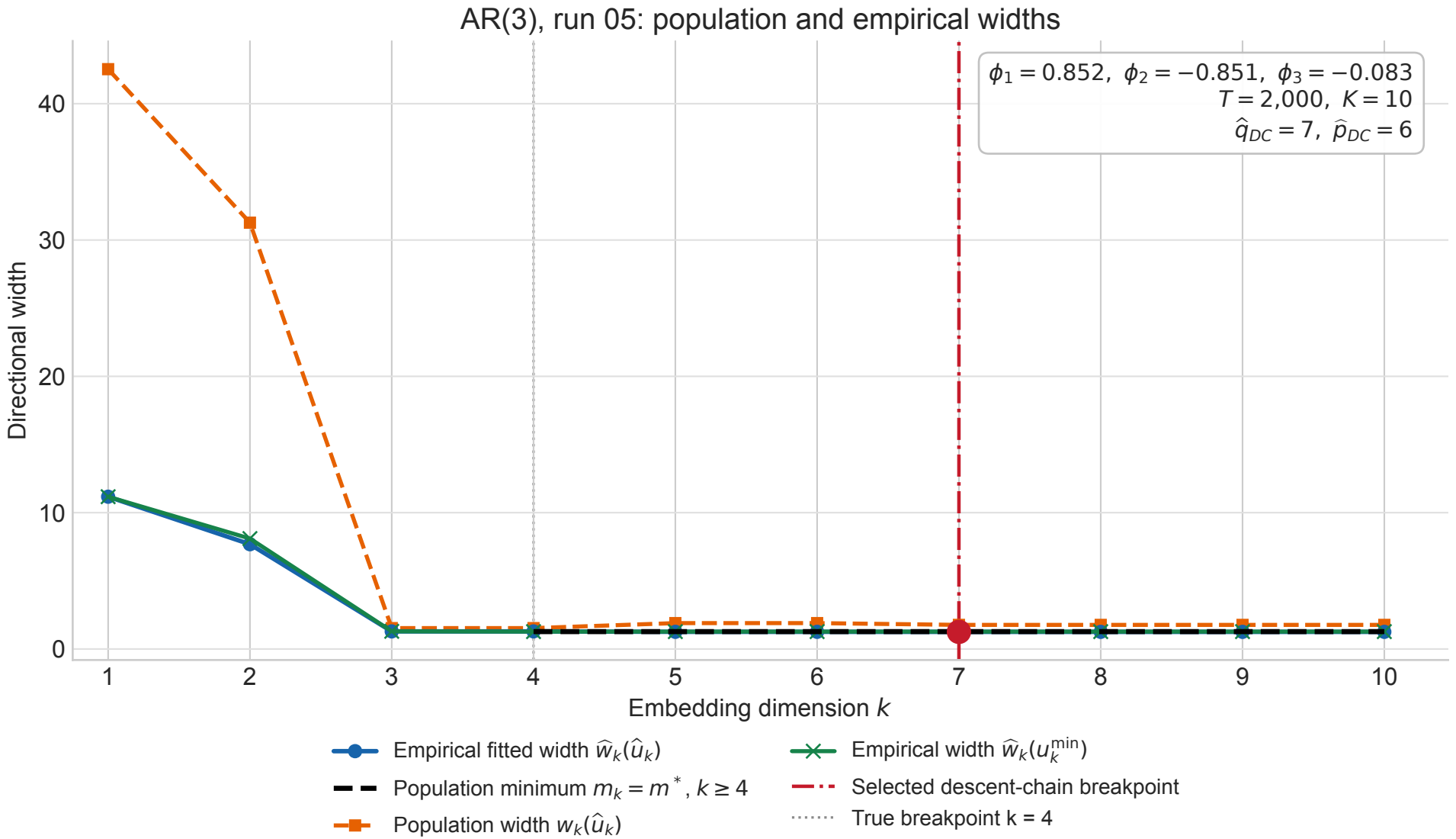
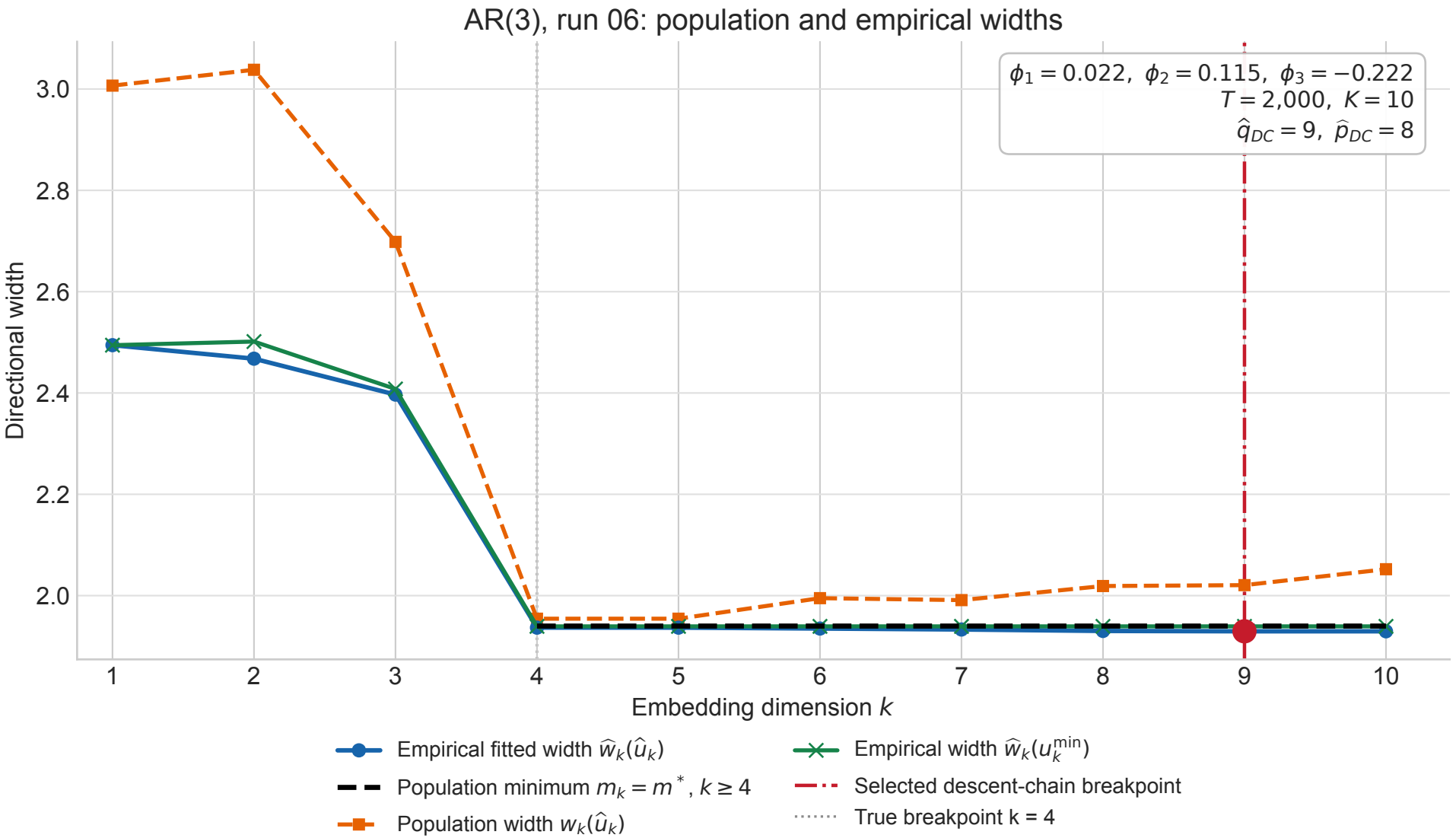

AR(3), run 01: population and empirical widths
$\phi_1 = -0.783,\ \phi_2 = 0.010,\ \phi_3 = 0.602$
$T = 2{,}000,\ K = 10$
$\hat{q}_{DC} = 5,\ \hat{p}_{DC} = 4$
Directional width
0
5
10
15
20
25
30
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\hat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\hat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ

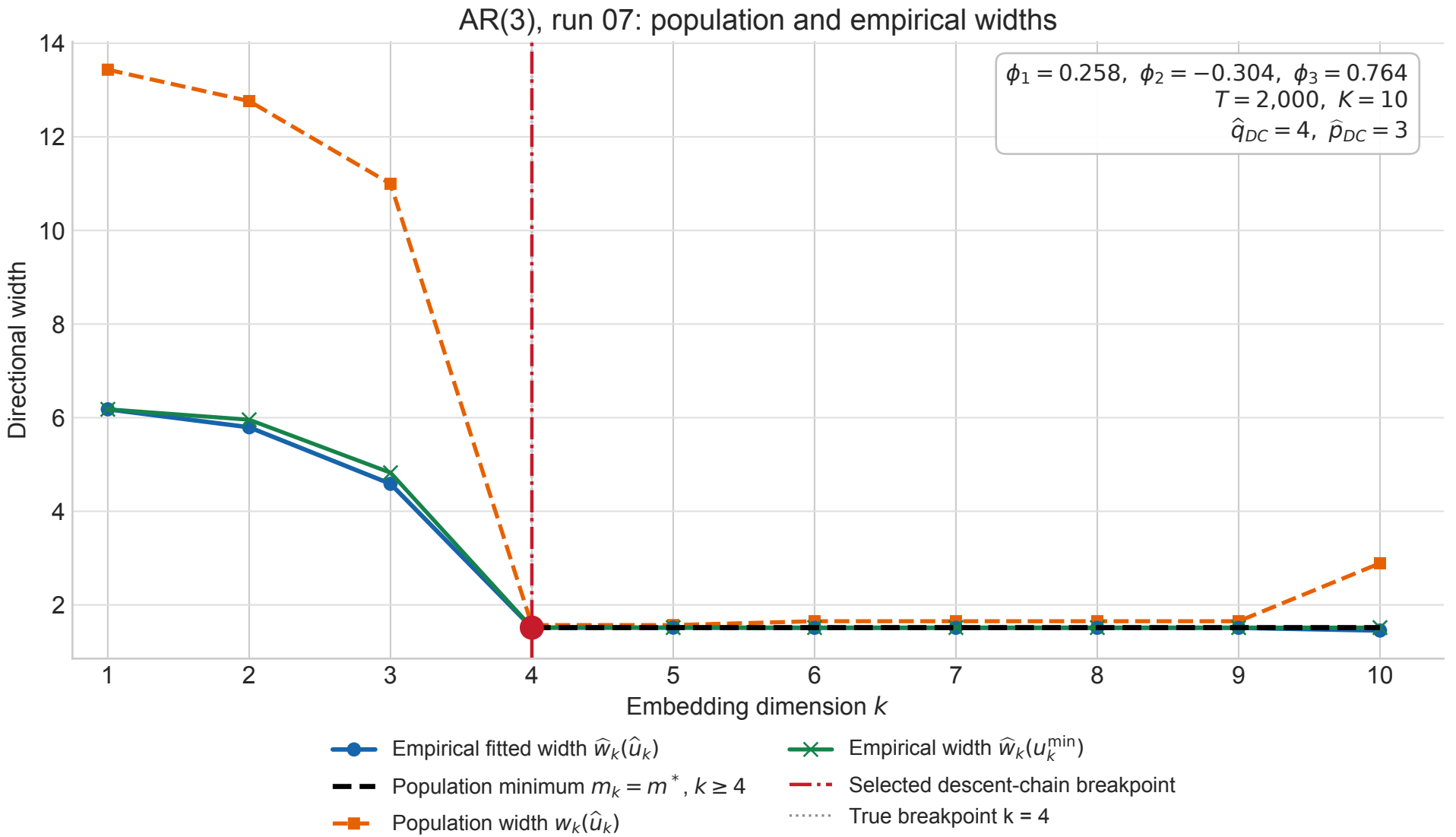

AR(3), run 02: population and empirical widths
$\phi_1 = 1.542,\ \phi_2 = -0.889,\ \phi_3 = 0.163$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 3,\ \widehat{p}_{DC} = 2$
Directional width
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ

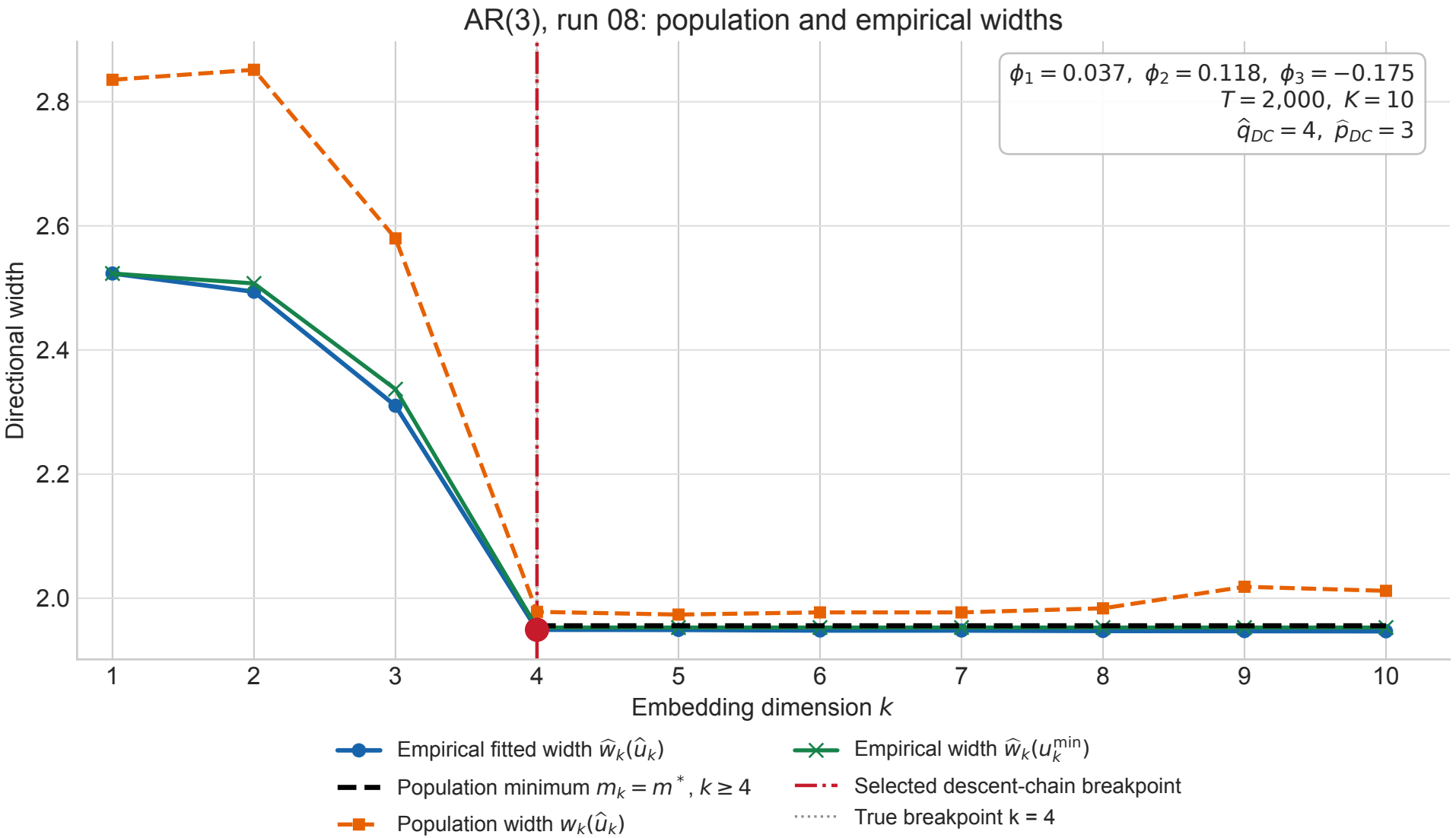

AR(3), run 03: population and empirical widths
$\phi_1 = 1.984,\ \phi_2 = -1.288,\ \phi_3 = 0.163$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 8,\ \widehat{p}_{DC} = 7$
Directional width
0
100
200
300
400
500
600
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ

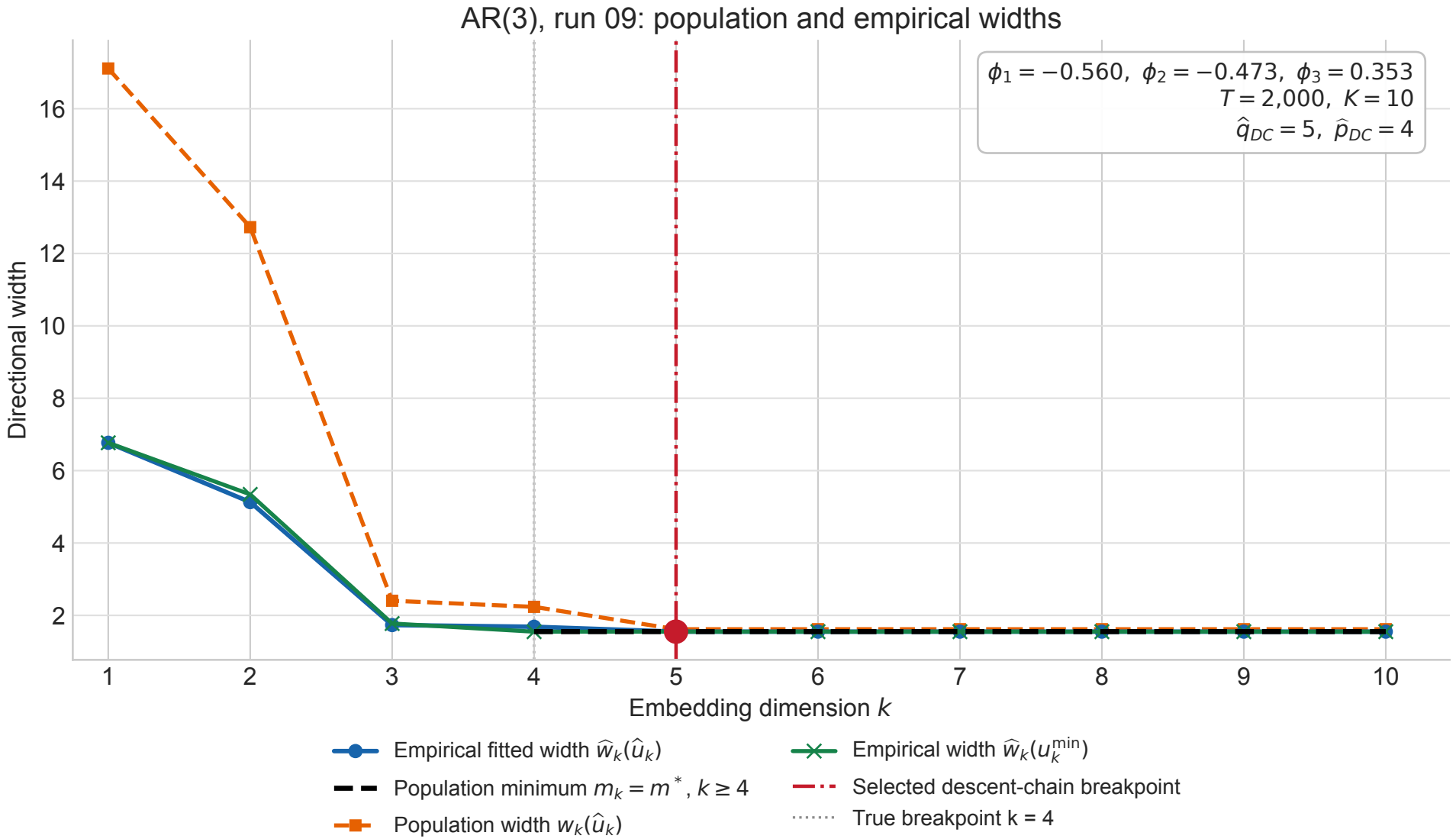

AR(3), run 04: population and empirical widths
$\phi_1 = -1.018,\ \phi_2 = 0.521,\ \phi_3 = 0.780$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 4,\ \widehat{p}_{DC} = 3$
Directional width
0
20
40
60
80
100
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ

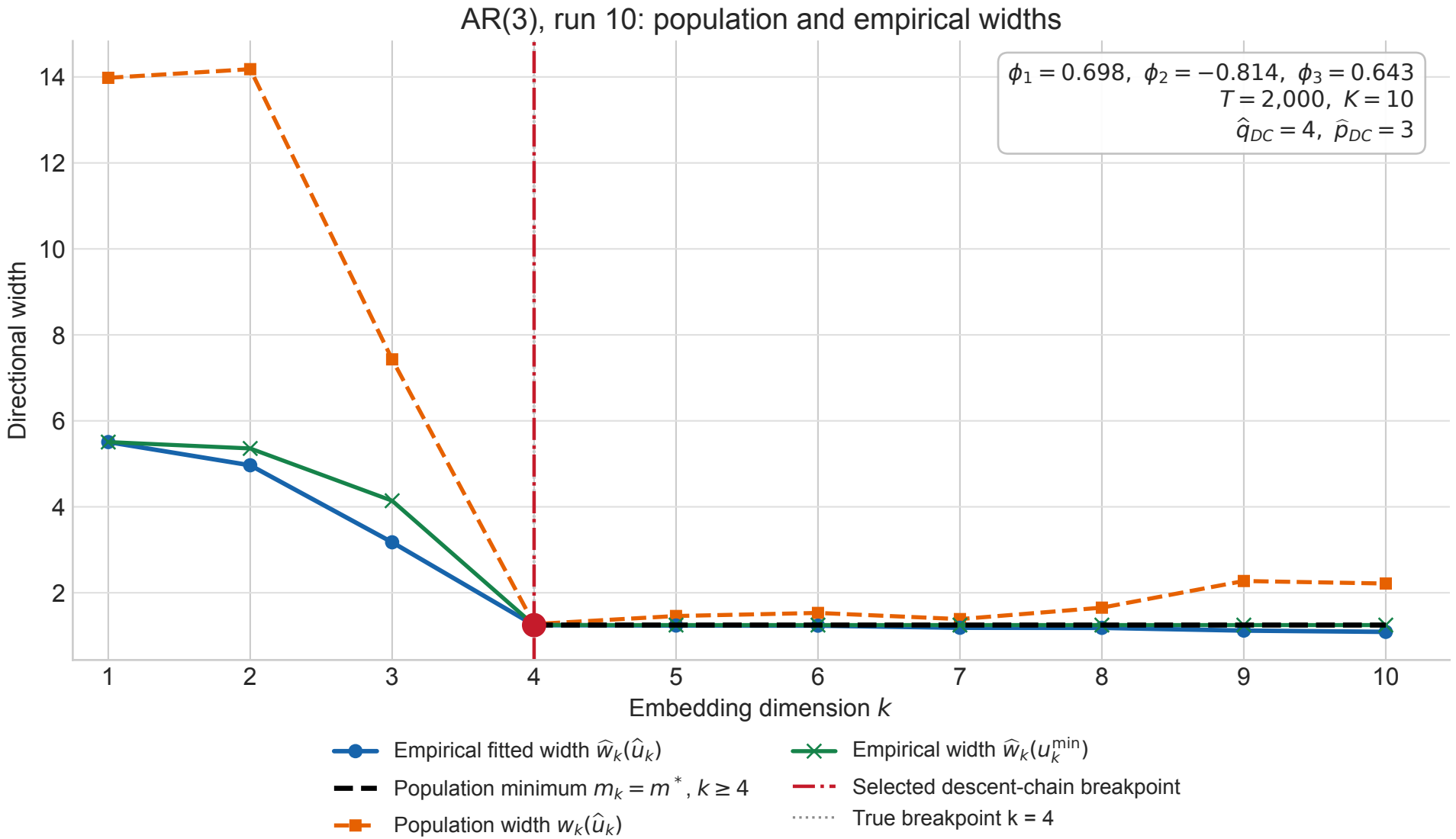

AR(3), run 05: population and empirical widths
$\phi_1 = 0.852,\ \phi_2 = -0.851,\ \phi_3 = -0.083$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 7,\ \widehat{p}_{DC} = 6$
Directional width
0
10
20
30
40
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ


AR(3), run 06: population and empirical widths
$\phi_1 = 0.022,\ \phi_2 = 0.115,\ \phi_3 = -0.222$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 9,\ \widehat{p}_{DC} = 8$
Directional width
3.0
2.8
2.6
2.4
2.2
2.0
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ


AR(3), run 07: population and empirical widths
$\phi_1 = 0.258,\ \phi_2 = -0.304,\ \phi_3 = 0.764$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 4,\ \widehat{p}_{DC} = 3$
Directional width
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Population minimum $m_k = m^*,\ k \geq 4$
Selected descent-chain breakpoint
Population width $w_k(\widehat{u}_k)$
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ


AR(3), run 08: population and empirical widths
$\phi_1 = 0.037,\ \phi_2 = 0.118,\ \phi_3 = -0.175$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 4,\ \widehat{p}_{DC} = 3$
Directional width
2.8
2.6
2.4
2.2
2.0
1
2
3
4
5
6
7
8
9
10
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ


AR(3), run 09: population and empirical widths
$\phi_1 = -0.560,\ \phi_2 = -0.473,\ \phi_3 = 0.353$
$T = 2{,}000,\ K = 10$
$\hat{q}_{DC} = 5,\ \hat{p}_{DC} = 4$
Directional width
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\hat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\hat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ


AR(3), run 10: population and empirical widths
$\phi_1 = 0.698,\ \phi_2 = -0.814,\ \phi_3 = 0.643$
$T = 2{,}000,\ K = 10$
$\widehat{q}_{DC} = 4,\ \widehat{p}_{DC} = 3$
Directional width
Embedding dimension k
Empirical fitted width $\widehat{w}_k(\widehat{u}_k)$
Population minimum $m_k = m^*,\ k \geq 4$
Population width $w_k(\widehat{u}_k)$
Empirical width $\widehat{w}_k(u_k^{\min})$
Selected descent-chain breakpoint
True breakpoint k = 4
Padded-start projected gradient descent: 3 starts, 500 iterations, step 0.2 × 0.99ⁿ